

1. Special Signals

As it turns out, there are quite a few special signals in the world of signal processing. For each of the following, plot the signal on the domain $[-4, 4]$. Then, answer the following questions:

- i: Is the signal continuous or discrete?
- ii: Is the signal even? (i.e. $x(t) = x(-t) \forall t$, or $x[n] = x[-n] \forall n$)
- iii: Is the signal odd? (i.e. $x(t) = -x(-t) \forall t$, or $x[n] = -x[-n] \forall n$)

- (a) Heaviside Step Function: $f : \mathbb{R} \rightarrow \mathbb{R}$, where $\forall t \in \mathbb{R}, f(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$

Answer: The plot of the Heaviside Step Function described is as follows.

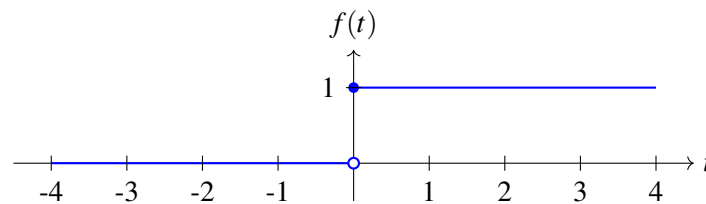
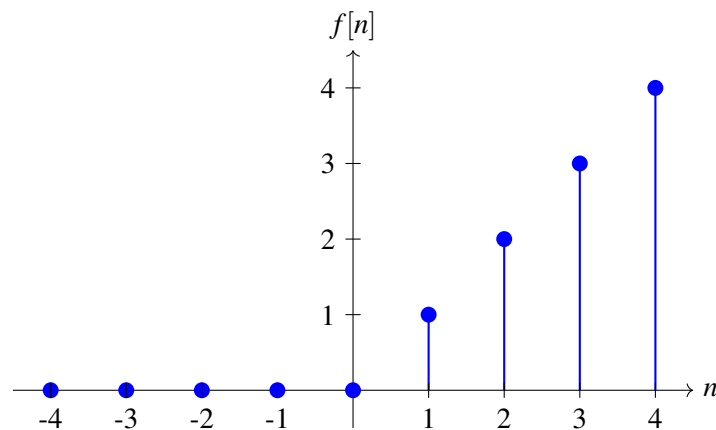


Figure 1: Heaviside Step Function $f(t)$

Because t is given as a real value, the signal should be continuous. Additionally, there is no symmetry about the y-axis or the origin, so the signal is neither even nor odd.

- (b) Ramp Function: $f : \mathbb{Z} \rightarrow \mathbb{R}$, where $\forall n \in \mathbb{Z}, f[n] = \begin{cases} 0 & \text{if } n < 0 \\ n & \text{if } n \geq 0 \end{cases}$

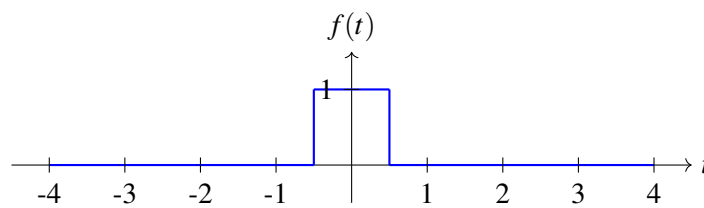
Answer: The plot of the Ramp Function is as follows.

Figure 2: Discrete Ramp Function $f[n]$

Because n is given as an integer, the signal should be discrete. Additionally, there is no symmetry about the y-axis or the origin, so the signal is neither even nor odd.

- (c) Rectangular Function: $f : \mathbb{R} \rightarrow \mathbb{R}$, where $\forall t \in \mathbb{R}, f(t) = \begin{cases} 1 & \text{if } |t| \leq 0.5 \\ 0 & \text{if } |t| > 0.5 \end{cases}$

Answer: The plot of the Rectangular Function described is as follows.

Figure 3: Continuous-Time Rectangle Function $f(t)$

Because t is given as a real value, the signal should be continuous. Observe in this case that there is symmetry across the y-axis, so the function is even, but there is no symmetry across the origin, so the function is not odd.

- (d) Kronecker Delta Function: $\delta : \mathbb{Z} \rightarrow \mathbb{R}$, where $\forall n \in \mathbb{Z}, \delta[n] = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \neq 0 \end{cases}$

Answer: The plot of the Kronecker Delta Function is as follows.

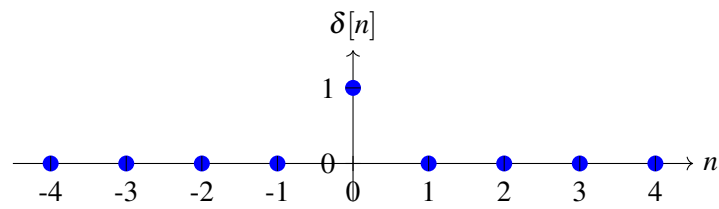


Figure 4: Discrete Kronecker Delta Function $\delta[n]$

Because n is given as an integer, the signal should be discrete. Observe in this case that $f[n] = f[-n] = 0$ for all $n \neq 0$, so the function is even. However, $f[0] = 1 \neq -1 = -f[-0]$, so it is not odd.

2. Interacting Discrete-Time Signals

Signals can be thought as vectors! Let's see how this is useful. Let the DT ramp signal $f[n]$ be defined as:

$$f[n] = \begin{cases} 0, & n < 0 \\ n, & n \geq 0 \end{cases}$$

Now define another signal:

$$g[n] = \begin{cases} 2, & n = -2, -1, 0, 1 \\ 0, & \text{otherwise} \end{cases}$$

For each question below we only consider the interval $-4 \leq n \leq 4$

(a) Sketch the signal $g[n]$

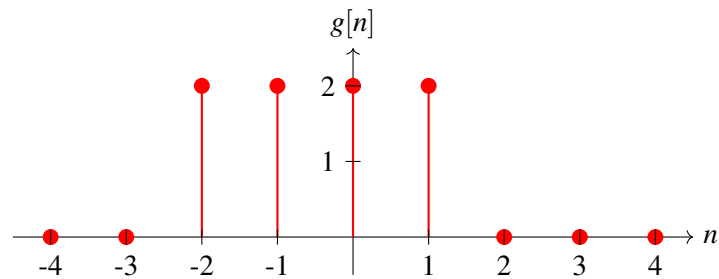
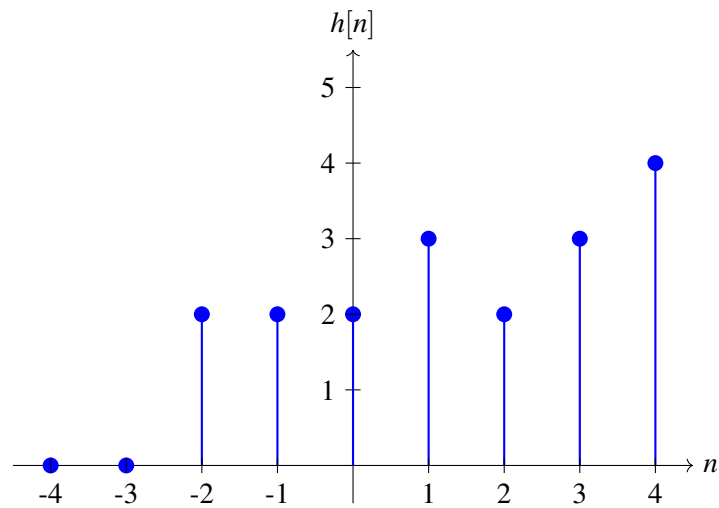
(b) Find and sketch a new signal $h[n] = f[n] + g[n]$

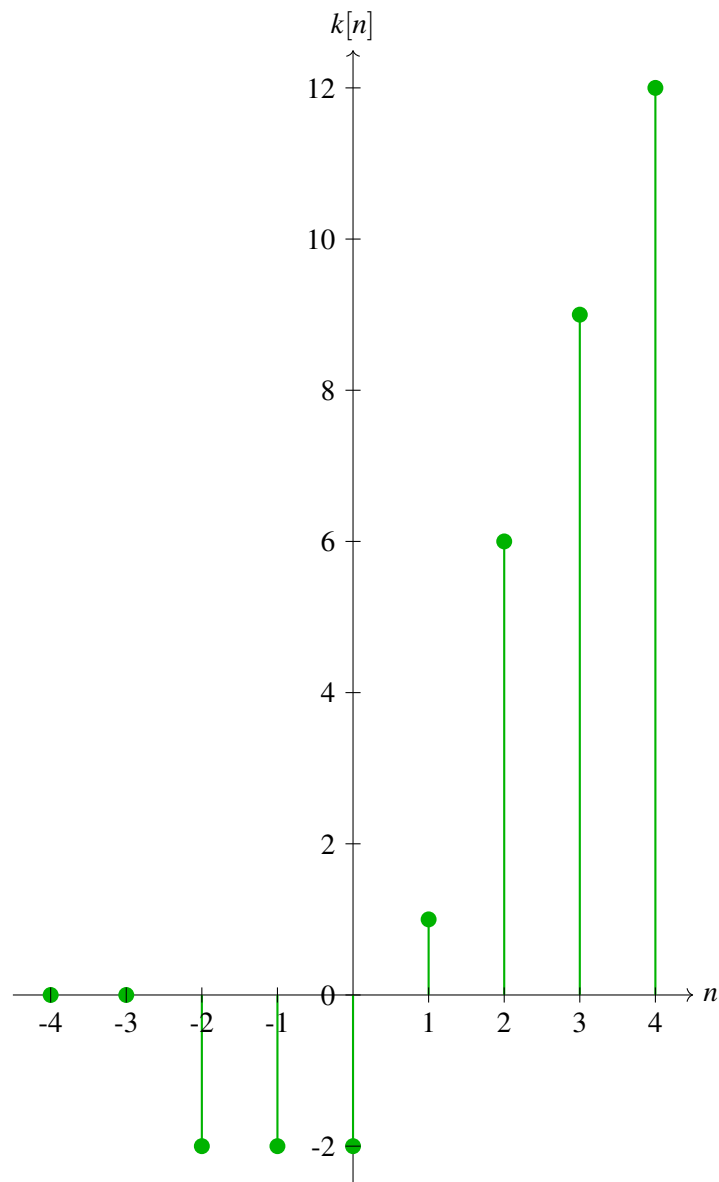
(c) Find and sketch another new signal $k[n] = 3f[n] - g[n]$

Answer:

We compute each signal for $n \in [-4, 4]$:

n	$f[n]$	$g[n]$	$h[n] = f[n] + g[n]$	$k[n] = 3f[n] - g[n]$
-4	0	0	0	0
-3	0	0	0	0
-2	0	2	2	-2
-1	0	2	2	-2
0	0	2	2	-2
1	1	2	3	1
2	2	0	2	6
3	3	0	3	9
4	4	0	4	12

Figure 5: Plot of $g[n]$ Figure 6: Plot of $h[n] = f[n] + g[n]$

Figure 7: Plot of $k[n] = 3f[n] - g[n]$

3. Inner Product Properties

For this question, we will verify our definition of the Euclidean inner product in Cartesian coordinates

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + \dots + x_n y_n, \quad \text{for any } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

indeed satisfies the key properties required for all inner products for the 2-dimensional case. Suppose $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^2$ for the following parts (there is a slightly different definition for complex vectors):

(a) Show symmetry: $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$.

Answer: This is seen by direct expansion:

Let $x_i, y_i \in \mathbb{R}$, then

$$\begin{aligned} \left\langle \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right\rangle &= x_1 \cdot y_1 + x_2 \cdot y_2 \\ &= y_1 \cdot x_1 + y_2 \cdot x_2 \\ &= \left\langle \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\rangle \end{aligned}$$

(b) Show linearity: $\langle \mathbf{x}, c\mathbf{y} + d\mathbf{z} \rangle = c\langle \mathbf{x}, \mathbf{y} \rangle + d\langle \mathbf{x}, \mathbf{z} \rangle$, where $c, d \in \mathbb{R}$ are real numbers.

Answer: This is accomplished through a direct expansion:

$$\begin{aligned} \left\langle \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, c \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} + d \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \right\rangle &= \left\langle \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \begin{bmatrix} cy_1 + dz_1 \\ cy_2 + dz_2 \end{bmatrix} \right\rangle \\ &= x_1(cy_1 + dz_1) + x_2(cy_2 + dz_2) \\ &= c(x_1 y_1 + x_2 y_2) + d(x_1 z_1 + x_2 z_2) \\ &= c \left\langle \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right\rangle + d \left\langle \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \right\rangle \\ &= c\langle \mathbf{x}, \mathbf{y} \rangle + d\langle \mathbf{x}, \mathbf{z} \rangle \end{aligned}$$

(c) Show non-negativity: $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$, with equality if and only if $\mathbf{x} = \mathbf{0}$.

Answer: This part requires just a bit more thought beyond a direct expansion of $\langle \mathbf{x}, \mathbf{x} \rangle$, but we first recognize that this inner product is the definition of the norm (or length) of $\mathbf{x}$. So it is at least intuitive that a length of some vector (squared) cannot be negative:

$$\begin{aligned}\langle \mathbf{x}, \mathbf{x} \rangle &= \left\langle \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\rangle \\ &= x_1^2 + x_2^2\end{aligned}$$

From this result we notice if either x_1 or x_2 are nonzero (even negative) values, then the inner product HAS to be positive. The only case in which the inner product $\langle \mathbf{x}, \mathbf{x} \rangle$ is identically zero is when both $x_1 = 0$ AND $x_2 = 0$, which verifies the final part of the property: $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ ONLY IF $\mathbf{x} = \mathbf{0}$.

As a bonus, suppose we re-label our vector components $x_1 = a$ and $x_2 = b$.

Then we see $\langle \mathbf{x}, \mathbf{x} \rangle = c^2 = a^2 + b^2$, which is the Pythagorean theorem!

This verifies that $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = c$ can be geometrically understood as the length of vector $\mathbf{x}$.